

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. BSC-09A: Statistical Inference, Sampling and Design of Experiments

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) Distinguish between sampling and complete enumeration. Explain the advantages of sampling over complete enumeration. [7]
 - (b) Explain the concept of a sampling frame. How is it useful in selecting a sample? [7]
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Question 2.

- (a) Describe stratified random sampling. Write its advantages over simple random sampling and give the formula for the estimate of the population mean. [7]
 - (b) Describe cluster sampling and two-stage sampling with examples. [7]
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Question 3.

- (a) What do you mean by design of experiment? Explain the principles of randomisation and replication. [7]
 - (b) Describe the analysis of variance table for a Completely Randomised Design (CRD). [7]
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Question 4.

- (a) Describe the Latin Square Design (LSD) and derive its analysis of variance table. [7]
 - (b) Describe the Randomised Block Design (RBD) along with its analysis of variance table. [7]
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Question 5.

- (a) Explain testing of hypothesis. In this context, describe the two kinds of errors (Type I and Type II) and the concept of level of significance. [7]
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- (b) Explain two important uses of the t -test in statistics. [7]
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Question 6.

- (a) Describe the various uses of the F -test. [7]
- (b) Explain proportional allocation and optimum allocation in stratified sampling. Derive the formula for optimum allocation. [7]
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Question 7.

- (a) Give an example where an experiment is not properly designed and explain how proper design of experiments can remedy such deficiencies. [7]
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Question 8. Write short notes on any **two** of the following: [14]

- (a) Systematic sampling
- (b) Factorial experiments
- (c) F -test
- (d) Multistage sampling